CAREEM BUILDERS ’26: HACK TO THE FUTURE

Payd — Standout Innovation Blueprint

Four Judge-Winning Architectural Extensions: Bedrock AgentCore Tracing, Aani Instant QR, Zero-Tap Ride Splitter & Arabizi Slang Intelligence

LEAD ENGINEER & AI	TARGET HACKATHON	TECH PILLARS	REGIONAL ANCHOR
Manan Arora (AI & Data Science)	**Careem Builders ’26 (Nov 14)**	**Bedrock AgentCore + Careem MCP**	**AED (100 fils) • CBUAE / FTA**

Executive Strategic Intent: While basic expense splitters wait for user inputs and perform approximate arithmetic, **Payd** evolves into an **Autonomous Financial Concierge**. By interfacing directly with **Careem MCP** and **Amazon Bedrock AgentCore**, Payd introduces four breakthrough capabilities designed to exceed the hackathon judging rubric: transparent agent reasoning, instant UAE Central Bank (Aani) QR settlement, proactive ride split triggers, and Khaleeji bilingual slang comprehension. All calculations strictly enforce Payd's five financial invariants (integer minor units in fils, zero float math, and deterministic remainder allocations).

Careem & AWS Scoring Alignment Matrix

How each proposed standout innovation maps directly to Careem's hackathon evaluation criteria:

Evaluation Dimension	Typical Hackathon Project	Payd's Standout Implementation	Judge Impact
AWS Bedrock Usage	Basic prompt wrapper calling Bedrock API.	**Live Cognitive Trace** showing ReAct loops, tool planning, and memory execution.	■■■■■■■■■■ Direct AWS rubric fit.
Careem MCP Utilization	Passive read of static ride data.	**Proactive Zero-Tap Trigger:** Intercepts ride finish & food delivery webhooks for 1-tap splits.	■■■■■■■■■■ Core Super-App synergy.
UAE Regional Context	Generic currency switcher ($ / € / AED).	**Aani Instant QR** + **FTA 5% VAT** + **Khaleeji / Arabizi** voice intelligence.	■■■■■■■■■■ Culturally unassailable.
Live Pitch Resilience	High risk of API rate limits & Wi-Fi crashes.	**Dual-Engine Reliability:** 100% deterministic offline fallback & sample presets.	■■■■■■■■■■ Zero demo failure risk.

Innovation 1: Amazon Bedrock AgentCore Live Reasoning Trace

Most LLM projects present a black box: the user speaks, a spinner runs, and a result appears. For technical judges from AWS and Careem, this conceals the architecture. Payd introduces an interactive **Cognitive Reasoning Trace Drawer** powered by Amazon Bedrock AgentCore, revealing each intermediate step of the ReAct (Reasoning + Acting) loop in real time.

LIVE BEDROCK AGENTCORE REASONING TRACE (UI VISUALIZATION)

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[User Input (Voice)]: "Hey Careem, split our ride to DIFC and dinner with Rahul and Sarah."
  Step 1 [Cognitive Planning]: Bedrock identifies composite intent: `split_transport` + `split_dining`.
  Step 2 [Careem MCP Invocation]: Calls careem_get_last_trip() → returns Hala Ride AED 64.00 (6,400 fils).
  Step 3 [Careem MCP Invocation]: Calls careem_get_last_order(dineout) → returns Al Safadi AED 260.00 (26,000 fils).
  Step 4 [Bilateral Debt Query]: Calls payd_get_debt_balance(Rahul) → Rahul currently owes user AED 20.00.
  Step 5 [Deterministic Ledger Invariant]: Handoff to /lib/ledger/splits.ts: total AED 324.00 split 3 ways yields 10,800 fils each. Remainder = 0 fils.
  Step 6 [Careem Pay Synthesis]: Generates net settlement cards (Sarah: AED 108.00; Rahul: AED 88.00 factoring prior debt).
```

Key Architectural Advantages:

- **Rigorous Tool Boundary:** Bedrock reasons across user intent and decides which MCP tools to call. However, it never calculates monetary allocations. It passes the raw parameters to Payd's deterministic TypeScript engine.
- **Single-Turn Clarification:** If Bedrock detects missing entities (e.g. participant names not in contact list), it triggers a dedicated clarification state with one precise question before touching the database.
- **Zero Prompt Injection Risk:** Financial values are strictly parsed into Zod schemas before being accepted into the ledger calculation pipeline.

Innovation 2: CBUAE Aani Instant P2P Settlement QR & WhatsApp Loop

Recording an expense solves only half of the consumer problem. Settle-up friction remains high because users must switch banking apps, search for IBANs, or copy amounts. Payd closes this loop by integrating the **Central Bank of UAE (CBUAE) Aani payment rail** and **WhatsApp Instant Deep-Links**.

Aani Dynamic QR Generator	Viral WhatsApp Settlement Deep-Link
When a user taps 'Settle' in the Settlements tab, Payd dynamically renders an **Aani-compliant QR Code** encoding: • Recipient Mobile Alias / UAE Proxy ID • Exact Integer Minor Units (AED fils) • Settlement Transaction Reference Token Any UAE resident can open their banking app (Wio Bank, Liv, Emirates NBD) or Careem Pay, point their camera at the screen, and authorize the transfer in under 5 seconds.	Payd generates a pre-formatted, one-tap WhatsApp deep link: https://wa.me/[phone]?text=... *"Marhaba Rahul! Your share for Careem DineOut is AED 42.50. Settle instantly via Aani or Careem Pay: careem://pay/request?ref=sp129"* Creates an immediate viral distribution loop across WhatsApp groups.

Innovation 3: Careem Super-App "Zero-Tap Auto-Split" Event Simulator

Current expense trackers are entirely passive: users must remember to open the app hours later and manually enter what happened. With Careem MCP, Payd shifts from a **passive record-keeper** to a **proactive agent** that intercepts Careem life events in real time.

Stage	Trigger Event	Agent Autonomous Action	User Experience
1. Trip Finish	User steps out of a Careem Hala Taxi in Dubai Marina.	Careem MCP webhook emits `ride.completed` with fare: AED 48.50, destination, and timestamp.	Push notification on phone: *"Careem Ride Completed"*.
2. Spatial Matching	Agent checks user's active groups.	Agent correlates destination coordinates with 'Dubai Flatmates' apartment location and member calendars.	Background agent synthesis without user prompting.
3. Zero-Tap Pop-up	User opens phone screen.	Presents pre-calculated confirmation card: *"Split AED 48.50 with Rahul and Sarah?"*	**1-Tap Confirm:** User taps 'Confirm' ➔ Split recorded in fils.
4. Auto-Settlement	Confirmation card accepted.	Dispatches settlement requests over Careem Pay P2P API automatically.	Instant confirmation notification sent to all participants.

Innovation 4: Khaleeji & Arabizi Voice Intelligence Engine

Dubai is one of the most culturally diverse cities in the world, with over 200 nationalities and widespread code-switching between English, Khaleeji Arabic, and Arabizi (text chat Arabic utilizing numerals for sounds: 7 for ■, 3 for ■, 2 for ■). Standard English-only voice models fail when expats speak naturally.

Spoken Voice Phrase	Dialect / Format	Entity Extraction & Parsing	Ledger Invariant Action
"Yalla 7esab Al Safadi dinner split with Rahul"	Arabizi + English code-switch	Merchant: Al Safadi Category: Food & Dining Split: Equal between User & Rahul	Prompts amount if missing, or matches to recent receipt scan.
"Khalo split 150 dirhams for Careem groceries"	Khaleeji colloquial Arabic	Intent: `split_expense` Amount: 15,000 fils (AED 150.00) Category: Groceries	Stores currency as AED, calculates integer shares.
"Kam 3lay for the flat rent?"	Gulf Arabic query ("How much do I owe?")	Intent: `query_balance` Entity: Rent & Accommodation	Queries deterministic ledger balance; formats reply in bilingual Arabic/English.

3-Minute Hackathon Winning Pitch Script & Protocol

A second-by-second execution playbook for the 3-person team during judging rounds on November 14:

Timestamp	Role / Presenter	On-Screen Action	Spoken Narrative / Key Punchline
0:00 - 0:30 (30 sec)	**Presenter 1** (Manan - Lead)	Display Home Dashboard with live AED balance and seed groups.	*"Splitwise just introduced 10-second paywalls and caps in the UAE. We built Payd: an autonomous, voice-first financial concierge engineered for Dubai with zero-float integer math."*
0:30 - 1:15 (45 sec)	**Presenter 2** (Teammate 2 - AI)	Tap floating mic and say: *"Hey Careem, split 240 dirhams for dinner with Rahul and Sarah."*	Show **Bedrock AgentCore Execution Trace**: highlight tool invocation, entity parsing, and zero-float ledger splitting into 80.00 AED minor units.
1:15 - 1:55 (40 sec)	**Presenter 1** (Manan)	Tap **Payd Lens** ➔ Select *Al Safadi Restaurant* receipt.	Show instant Vision OCR extraction: 15-digit TRN, subtotal, and **5% UAE VAT**. Demonstrate itemized line items and proportional VAT distribution.
1:55 - 2:30 (35 sec)	**Presenter 3** (Teammate 3 - Product)	Open Settlements tab ➔ click **'Settle with Aani QR'**.	Hold phone to the judges: *"Judges, scan this QR code with your phone camera right now. It generates an instant Aani / Careem Pay settlement link."*
2:30 - 3:00 (30 sec)	**Full Team**	Show Native iOS Swift App running in Simulator side-by-side with Next.js PWA.	*"Payd is dual-platform, fully tested across 31 invariants, non-custodial under CBUAE regulations, and ready for deployment on Careem Super App today. Thank you!"*

Pre-Hackathon Readiness Checklist:
- **Codebase Categorization:** Dual-platform monorepo split into web/ (Next.js 15 PWA) and ios/ (SwiftUI + SwiftData).
- **Test Coverage:** 31 Vitest unit & acceptance tests passing (100% test pass rate across all financial invariants).
- **Careem & Bedrock Tool Stubs:** Mock MCP schemas and Bedrock client adapter ready for instant credential injection on Saturday.